

International Finance

Lecture V– March 17th

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IES FSV UK – Spring 2026

Moodle: <https://dl1.cuni.cz/course/view.php?id=4605>

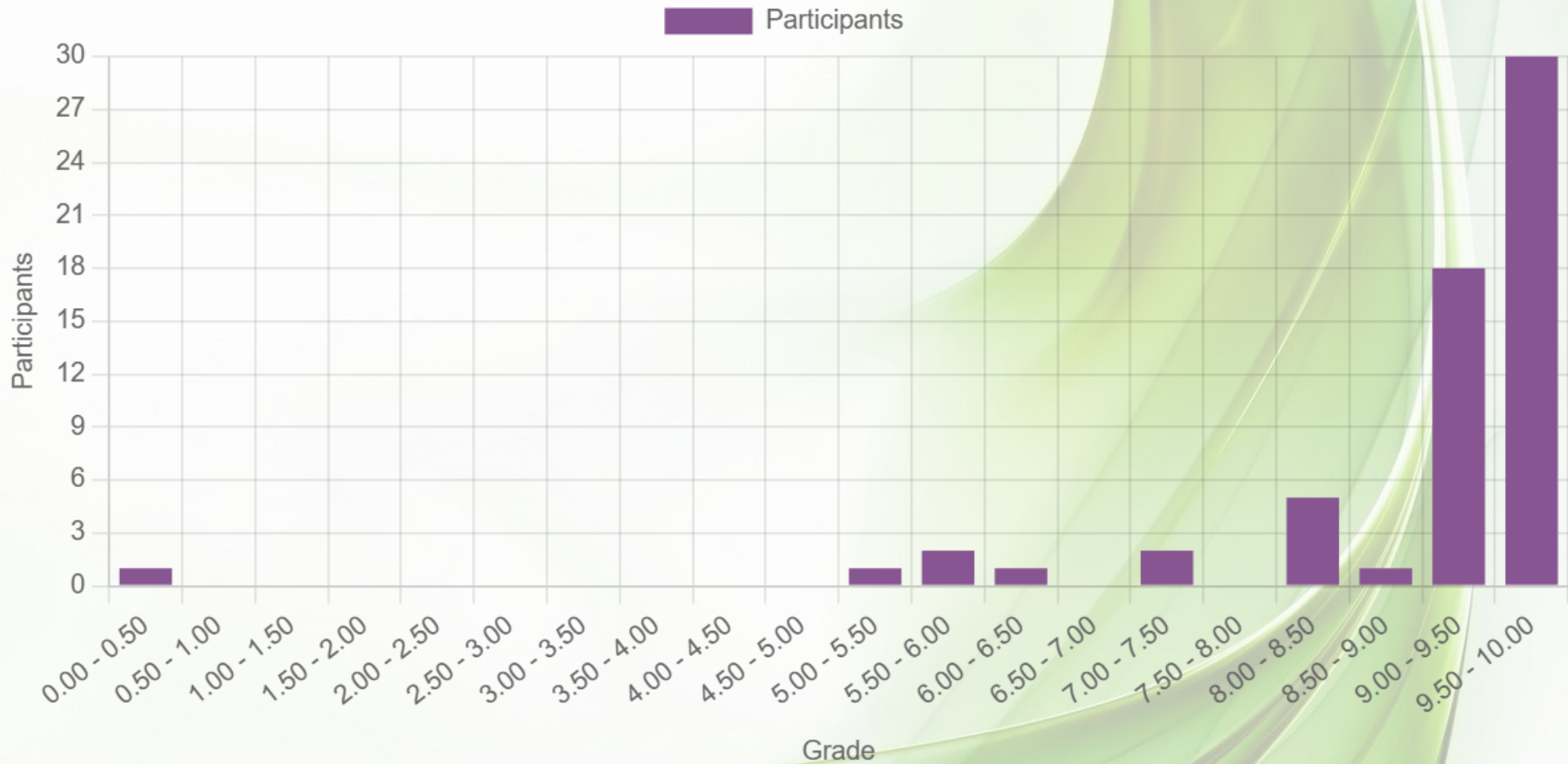
(International Finance JEB050/JPB335)

Course enrolment key: **PostUSD?2026**

Outline

- Course organization
 - Attendance, quiz, assignments (Teams + topics)
 - Quiz #5: will be open by tomorrow (Wednesday) midnight
 - The usual deadline: next Monday midnight
- Covered interest rate parity
 - Forward exchange rates
 - Using forwards as predictors?
- Exchange rates in the long run
 - Purchasing power parity
 - Absolute version
 - Relative version
 - Empirical issues

Quiz #4



Average: 9.03

70 attempts/65 participants

Reading for the Next Week:

- Krugman & Obstfeld & Melitz: International Economics (chapter 16)
- Taylor & Taylor (2004): The Purchasing Power Parity Debate

Course Outline (Changes Possible)

1. Introduction to international finance. Foreign exchange market: main actors, their role, types of interactions. Brief outline of the history of international finance. (February 17th)
2. International transfers/payments. Channels, related instruments. (February 24th)
3. Balance of payments structure and national income accounting for an open economy. International investment position (March 3rd)
4. Introduction to exchange rate determination: asset approach. **Covered and uncovered interest rate parity.** (March 10th)
5. **Money, interest rates and exchange rates. Long-run aspects of exchange rate determination. Purchasing power parity. Overshooting.** (March 17th)
6. Prices and exchange rates continued: Fischer effect. Exchange rates and competitiveness: real (effective) exchange rates, Balassa-Samuelson effect. Application on the nominal and real convergence in the EU (March 24th)
7. Introduction to the AA-DD model. Temporary and permanent changes in fiscal and monetary policies. (March 31st)
8. The AA-DD model: XX curve. Application on policy analysis. (April 7th)
9. Policy options for reaching internal and external balance. Swan diagram. Fixed exchange rates: macroeconomic implications. Interventions and sterilisation. Policy trilemma. (April 14th)
10. **Monetary integration: costs and benefits. Monetary integration: effects – empirical research.** (April 21st)
11. A brief introduction to the models of balance of payments crises. (April 28th)
12. Debts and (developing) countries. **Deadline for the team assignment/paper** (May 5th)
13. The future of IMS, role of the USD. Discussion of current topics. (May 12th)
14. **Reserved for a make-up class (possibly for April 21st). (May 19th)**

Back to Un/Covered Interest Parity and Forward Exchange



Exchange Rates and Interest Rates

- Two types of seemingly very similar relationships:
 - Uncovered interest rate parity (UIP)
 - Equilibrium relationship between spot exchange rate, expected exchange rate and interest rates
 - *Covered interest rate parity (CIP)*
 - *Arbitrage relationship between spot, forward exchange rate and interest rates*

Covered Interest Parity (CIP)

- Very similar to UIP, but this time we analyze forward markets

$$Deposit \cdot (1 + R_{CZK}) = Deposit \cdot \frac{1}{S_{CZK/\text{€}}} \cdot (1 + R_{\text{€}}) \cdot F_{CZK/\text{€}}$$

- But: *don't forget about possible transaction costs!*

- Forward rate: $F_{CZK/\text{€}} = S_{CZK/\text{€}} \cdot \frac{(1 + R_{CZK})}{(1 + R_{\text{€}})}$

- Or again simply: $F_{CZK/\text{€}} - S_{CZK/\text{€}}(\text{rel. change}) \cong R_{CZK} - R_{\text{€}}$

CNB: Current Forward Rates

Selected forward exchange rates

16 Mar 2026

EUR/CZK

maturity	forward points
3M	87.05
6M	164.50

USD/CZK

maturity	forward points
3M	-18.20
6M	-35.90

Forward points = Forward premium/discount

Forward premiums or discounts are usually quoted in **pips or points (1/100 of 1%)**

CNB: The forward exchange rate is calculated by **dividing the forward points by 1000** and adding the result to the spot rate

Vice versa: Calculating the **forward premium**:

$$FP = (R_{CZK} - R_{\text{€}}) \cdot S \cdot \frac{n}{360}$$

Converting to pips: **FP x 10000**.

S ... spot exchange rate

n ... number of days

$(I_F - I_D)$... interest rate differential

Source: Czech National Bank

<https://www.cnb.cz/en/financial-markets/foreign-exchange-market/forward-exchange-rates/forward-exchange-rates/index.html>

Empirical Issues

- There is strong empirical evidence that the covered interest parity condition holds for different foreign currency deposits issued within a single financial center.
- Indeed, currency traders often set the forward exchange rates they quote by looking at current interest rates and spot exchange rates and using the covered interest parity formula.
 - But, once differences in perceived risk are added, this is no longer necessarily true
 - E.g. If deposits are located in different countries.

Fama's Method

Fama (1984): Forward and spot exchange rate

$$F_t = E(S_{t+1}) + P_t$$

Where: $F_t = \ln f_t$ $S_{t+1} = \ln s_{t+1}$

$P_t = \text{premium}$

$$F_t - S_t = P_t + E(S_{t+1} - S_t)$$

- Consider regressions of $F_t - S_{t+1}$ and $S_{t+1} - S_t$ (observed at time t+1) on $F_t - S_t$ (observed at t)

$$S_{t+1} - S_t = \alpha_2 + \beta_2(F_t - S_t) + \epsilon_{2,t+1}$$

- If β_2 non-zero, then the forward rate has information about the spot rate to be observed at t+1.
 - And what if $\beta_2 = 1$?
 - In an efficient and rational exchange market, the deviation of β_2 from 1 is a direct measure of the variations in the premium of the forward rate

Predictive Power

Forward Premium Puzzle

- Fama (1984): Forward and spot exchange rate
- *There is a general consensus that forward exchange rates have little if any power as forecasts of future spot exchange rates. There is less agreement on whether forward rates contain time varying premiums.*
- *Conditional on the hypothesis that the forward market is efficient or rational, this paper finds that both components of forward rates vary through time. Moreover, **most of the variation in forward rates is variation in premium**, and the premium and expected future spot rate components of forward rates are negatively correlated.*

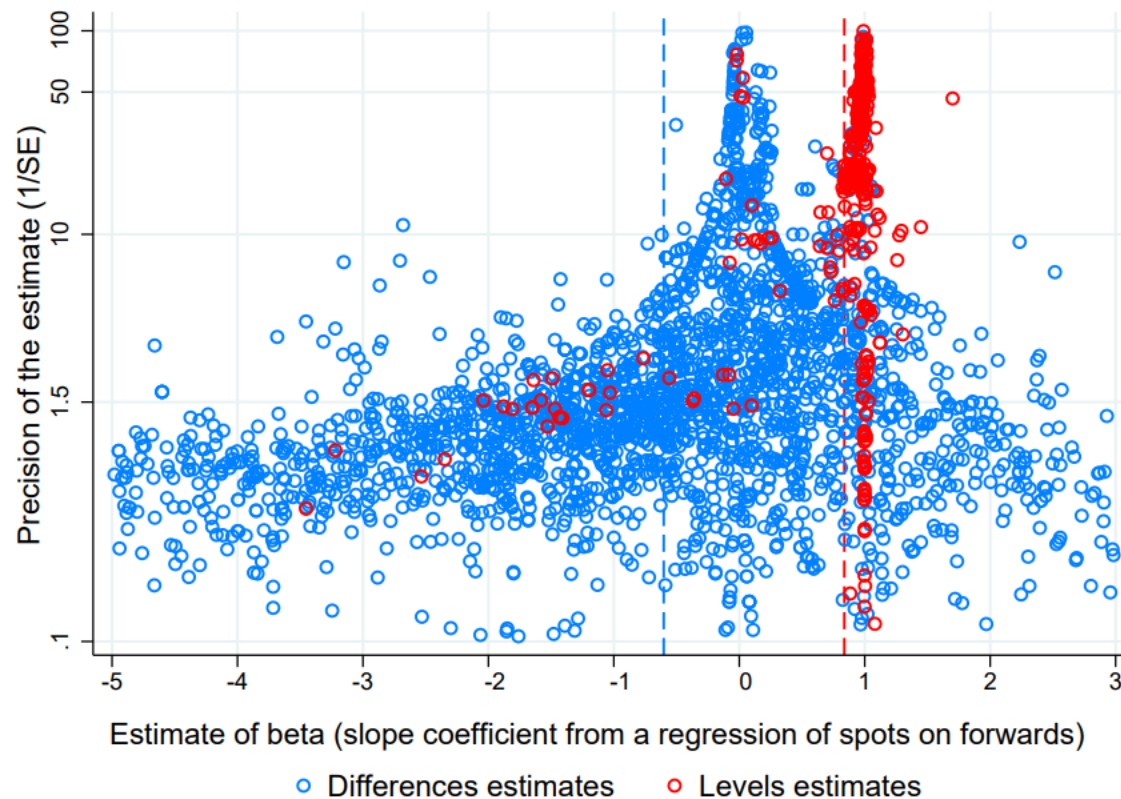
Some Other Claims

- Agmon & Amihud (1981): The forward exchange rate and the prediction of the future spot rate: Empirical evidence.
 - *...it does not perform better than the current spot rate in predicting the future spot rate for all the examined forecast leads.*
- Ha & Reddell (1998): What do forward interest and exchange rates tell us?
 - *Forward rates tell us very little about where the actual rate will be.*
- Meredith & Ma (2002): The forward premium puzzle revisited.
 - The forward premium is a notoriously poor predictor of exchange rate movements

A Bit Ecclectic Result

- Zigraiova, Havranek, Novak (2020): How puzzling is the forward premium puzzle? A meta-analysis
 - Scores of empirical studies report negative slope coefficients from regressions of spot rates on forward rates, which is inconsistent with the forward rate unbiasedness hypothesis
 - They collect 3,643 estimates from 91 research articles and using recently developed techniques investigate the effect of publication and misspecification biases on the reported results.
 - **Correcting for these biases** yields slope coefficients of 0.31 and 0.98 for developed and emerging currencies respectively, which implies that empirical evidence is **in line with the theoretical prediction for emerging economies** and less puzzling than commonly thought for developed economies.

Figure 1: *Are positive estimates underreported?*



Notes: In the absence of publication bias, the scatter plot should resemble an inverted funnel that is symmetrical around the most precise estimates. Red circles (darker in grayscale) represent estimates extracted from the levels equation (Equation 2); blue circles (lighter in grayscale) show estimates from the differences equation (Equation 3). Outlying observations are cut from the figure for ease of exposition but included in all statistical tests. The red dashed vertical line shows the mean estimate from the levels specification (0.84); the blue dashed vertical line shows the mean from the differences specification (−0.60).

Forwards (and the Puzzle)

Can they be used for predictions?

- A currency's forward rates should form unbiased predictors of future spot rates - under the assumptions of **risk neutrality** and **rational expectations**.
- Yes, there is **significant conflicting empirical evidence**

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Exchange Rates in the Long Run

Purchasing Power Parity

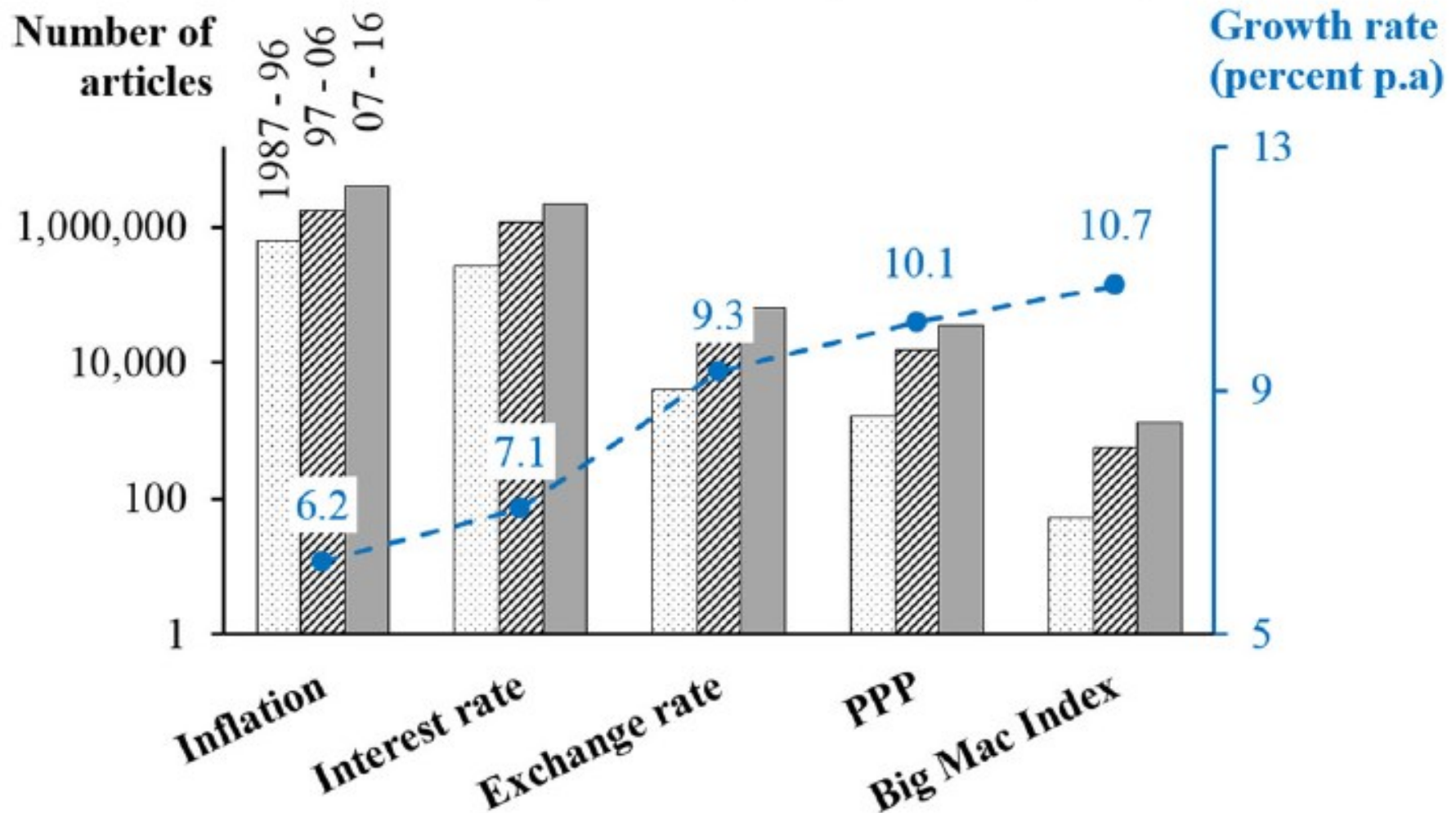
History of the Idea

- Origin of the logic of the idea: very old
 - University of Salamanca, 16th century
- Modern version: early 20th century
 - Post WWI discussions about approximately correct levels of exchange rates after the inflationary effects of WWI
 - **Gustav Cassel**
- Present (both in models and as intuitive framework) ever since:
 - Dornbusch and Krugman (1976) “Under the skin of any international economist lies a deep-seated belief in some variant of the PPP theory of the exchange rate.”
 - Rogoff (1996) “While few empirically literate economists take PPP seriously as a short-term proposition, most instinctively believe in some variant of purchasing power parity as an anchor for long-run real exchange rates.”

Gustav Cassel (1922):

- ***Our willingness to pay a certain price for foreign money must ultimately and essentially be due to the fact that this money possesses a purchasing power as against commodities and services in that country. On the other hand, when we offer so and so much of our own money, we are actually offering a purchasing power as against commodities and services in our own country.***
- ***Our valuation of a foreign currency in terms of our own, therefore, mainly depends on the relative purchasing power of the two currencies in their respective countries.***

Professional Interest in Economic Topics, search performed Aug 13, 2016



Purchasing Power Parity

- **Absolute PPP**

- A unit of currency should be able to buy the same basket of goods in one country as the equivalent amount of foreign currency, at the going exchange rate, can buy in a foreign country,
- The currency should have the same purchasing power at home and abroad

$$P_{CZ} = P_{GER} \cdot S_{CZK/EUR} \Rightarrow S_{CZK/EUR} = \frac{P_{CZ}}{P_{GER}}$$

- **Logic:** if PPP does not hold, we should observe large imbalance in the balance of trade, then the exchange rate would adjust back to PPP

$$S > S_{PPP} \Rightarrow X > M \Rightarrow S \downarrow$$

$$S < S_{PPP} \Rightarrow M > X \Rightarrow S \uparrow$$

Balance of Payments

Item	Credit	Debit	Macrostructure
Trade in goods			<i>Current Account</i>
Services			
Primary income			
Secondary income			
Capital account			<i>Capital Account</i>
Direct inv.			<i>Financial Account</i>
Portfolio inv.			
Other capital			
Errors & omissions			
Change in reserves			<i>Reserves</i>

Assumptions/Prerequisites

- Perfect validity of the PPP in its absolute form assumes:
 - **Perfect mobility** of goods/services (low transportation costs)
 - **Absence of significant qualitative differences** of the consumed commodities services
 - Absence of significant trade policy barriers
 - Absence of significant market entry/exit barriers
 - Absence of other motives for demanding/holding the particular currencies
- Additional “empirical” problem: different composition of consumption/GDP (a form of index number problem)

Reading for the This Week:

- Economist (2025) - Don't let Donald Trump see our Big Mac index
- Economist (2024): What Donald Trump can learn from the Big Mac index

Popularity: Simple “Concepts”

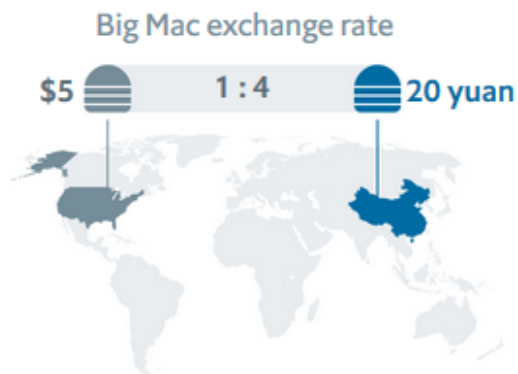
- Economist: the Big Mac index
 - <https://www.economist.com/big-mac-index>
 - “Invented” in 1986

How it works

Purchasing-power parity implies that exchange rates are determined by the value of goods that currencies can buy



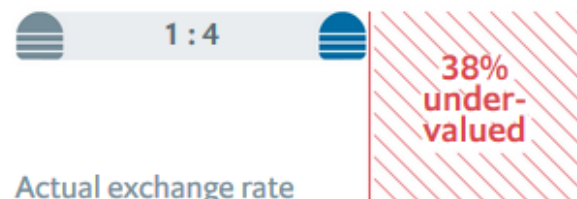
Differences in local prices – in our case, for Big Macs – can suggest what the exchange rate should be



Raw index /GDP-adjusted

Using burgernomics, we can estimate how much one currency is under- or over-valued relative to another

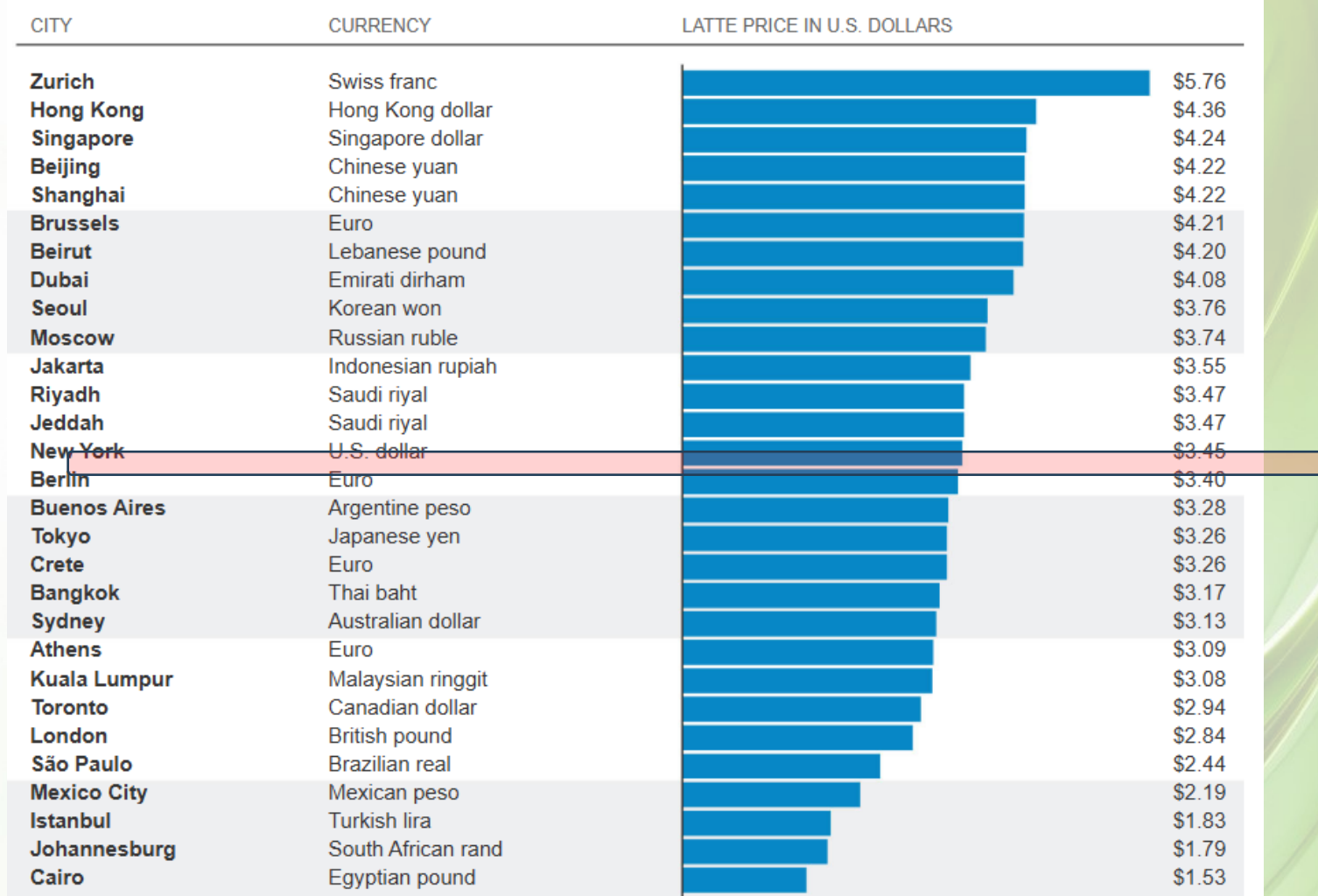
Big Mac exchange rate



Actual exchange rate



Of course, you can “pick your own...” Wall Street & Starbucks Latte (2017)...



Source: Staff reports

- Frankly, this case can be even more troublesome – why?

Pros and Cons

- Why should something like this work?
 - Very similar (next to identical) commodities
- And why not?
 - Tradability issues:
 - Can there really be an arbitrage between a latte in Paris and New York?
 - How about inputs? Many of them are local and non-tradable (labour costs, real estate...)
 - Identical physical features of a product do not necessarily imply identical market positioning....

1990: McDonald comes to Moscow



1990: McDonald comes to Moscow

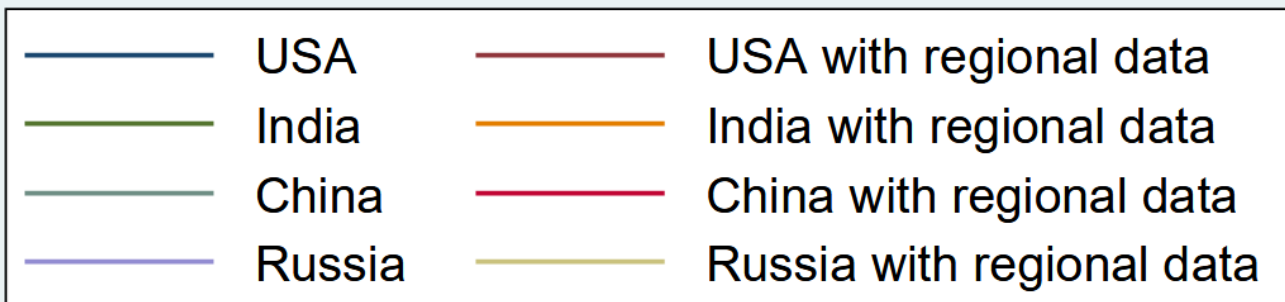
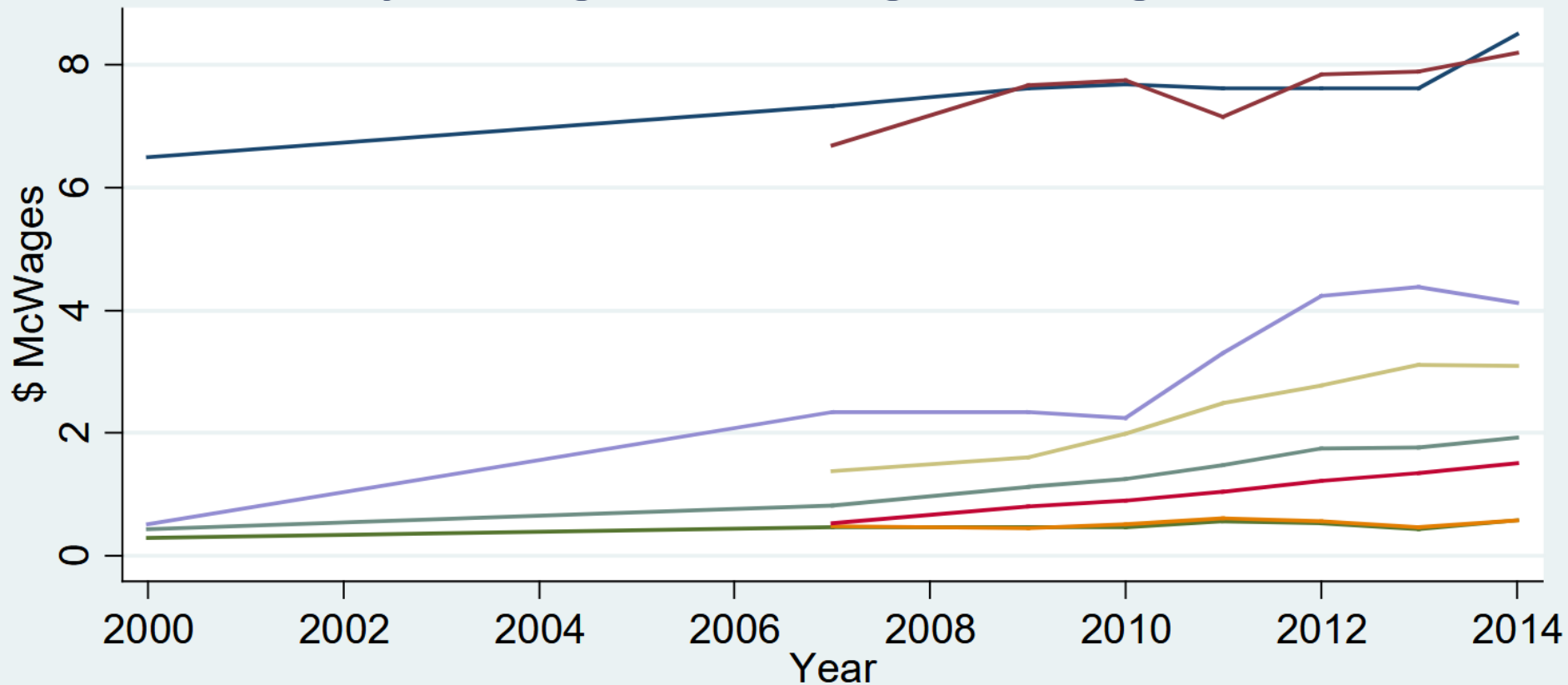


Source: <https://www.boredpanda.com/first-mcdonald-restaurant-opens-soviet-union-moscow-russia-1990/>

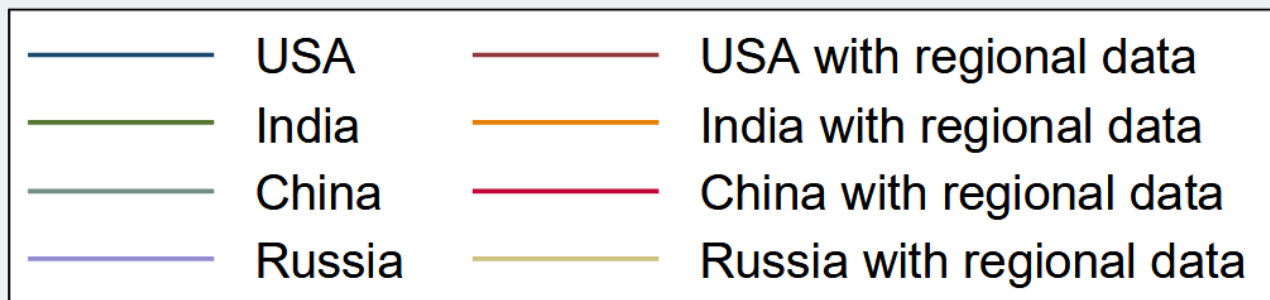
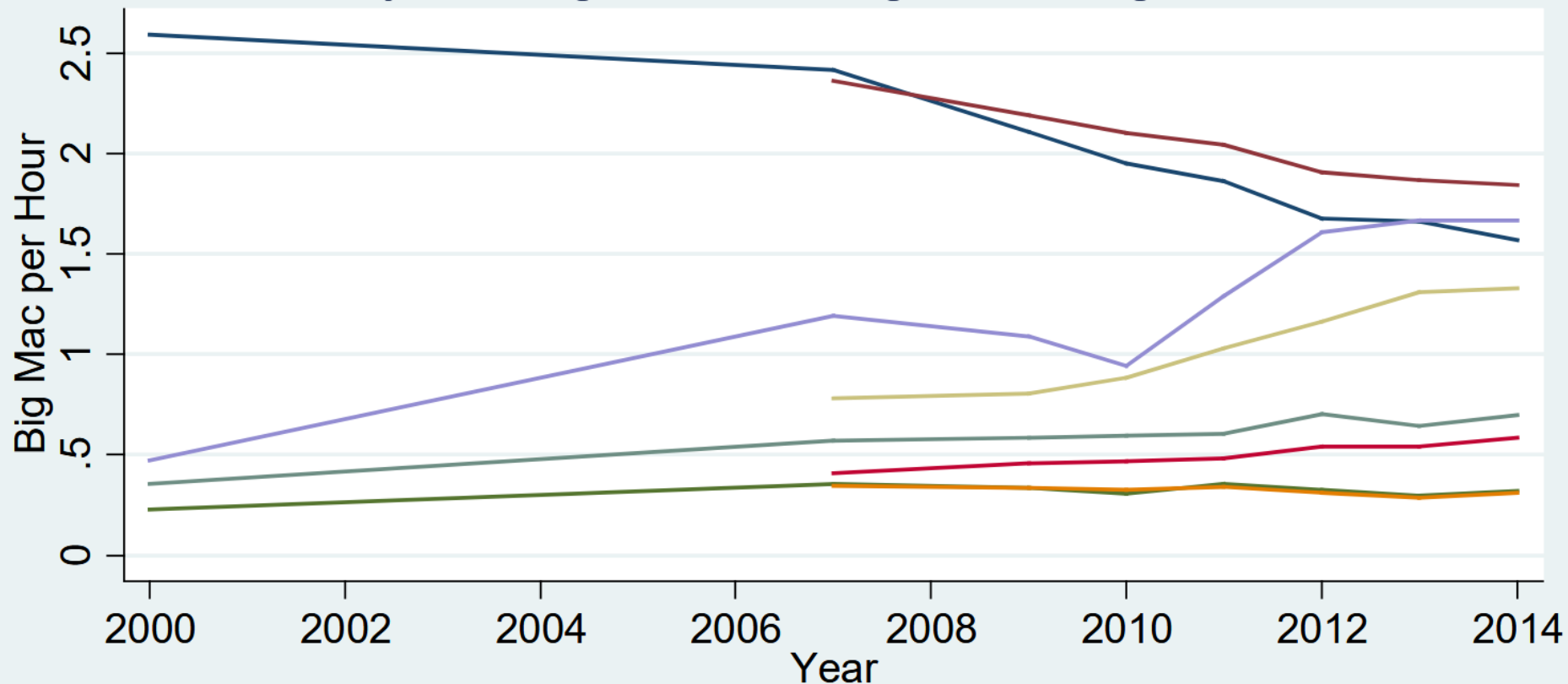
McWages

- Ashenfelter & Jurajda: Comparing Real Wages using McWages
- This can be a much better idea than the Big Mac parities
- Why? Much better comparability
 - Entry-level basic-crew jobs at McDonald's are virtually identical in terms of
 - skill input
 - technology (productivity)
 - hedonic job qualities
 - producing identical product in over 140 countries of the world.
 - Operations are monitored using 600-page Operations and Training Manual
 - McDs do not adjust technology
 - pay local market wage (MP)

Evolution of McWages: 2-city average vs. average with regional data

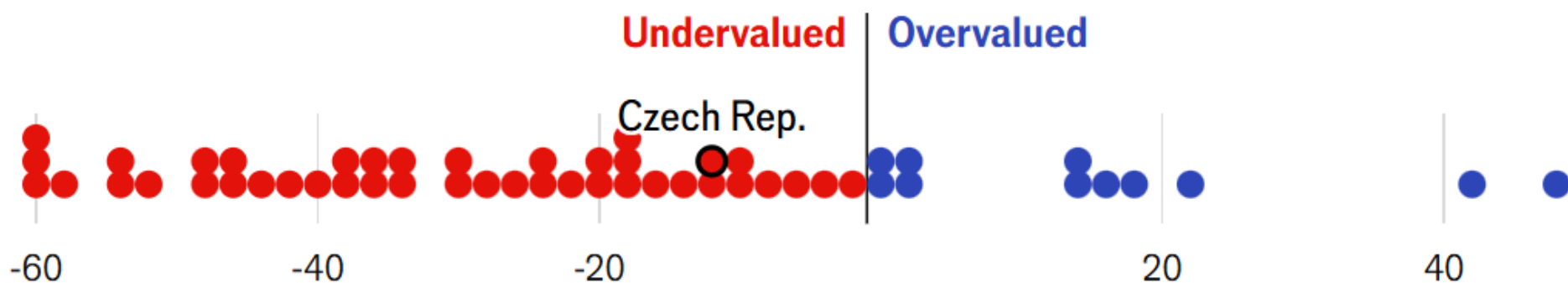


Evolution of BMPH: 2-city average vs. average with regional data



But: Let's Try to look at Big Macs....

Jan 2026 | **The Czech koruna is 10.2% undervalued against the US dollar**

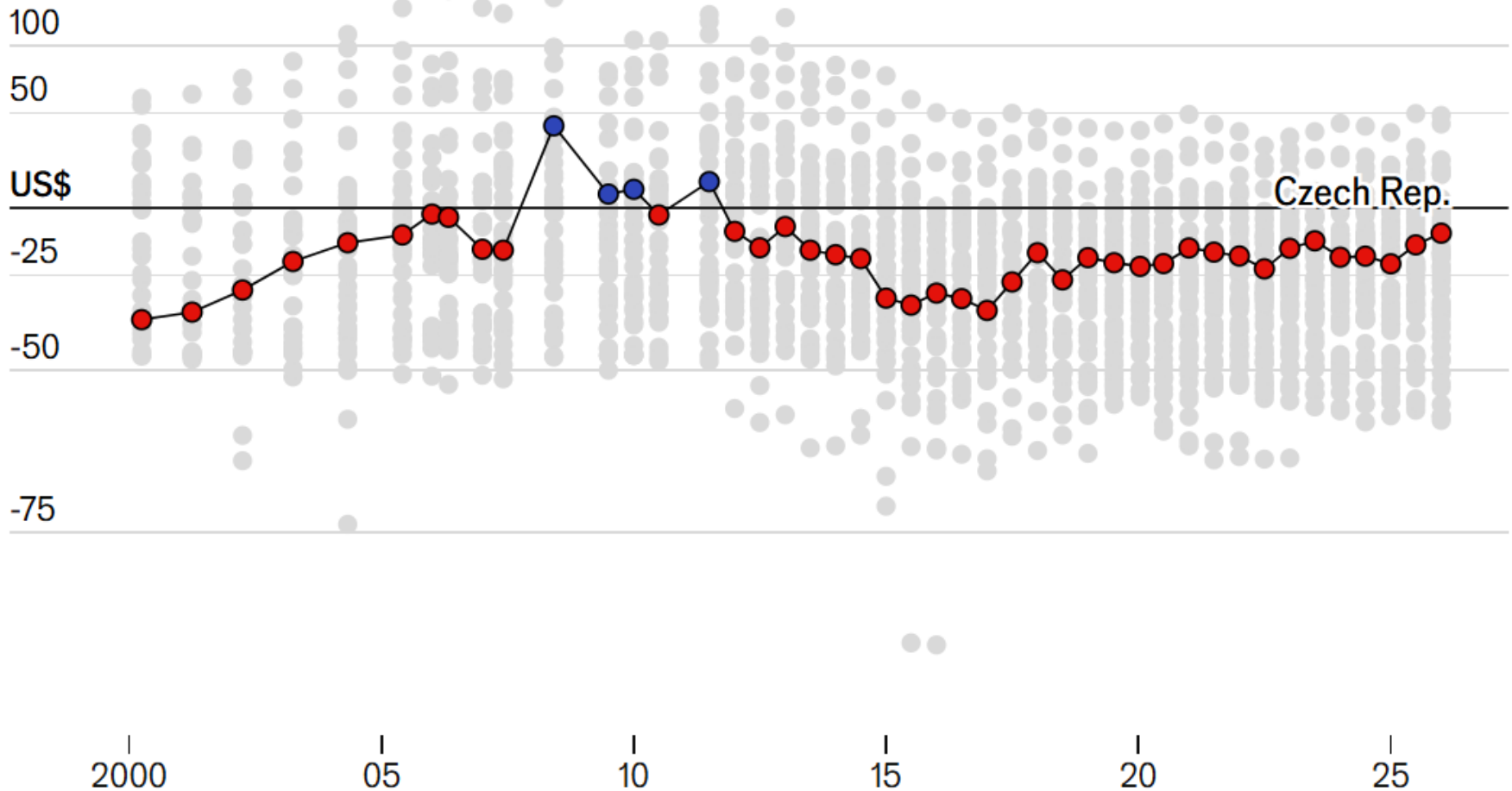


A Big Mac costs **115 koruna** in Czech Rep. and **US\$6.12** in **the United States**. The implied exchange rate is **18.79**. The difference between this and the actual exchange rate, **20.93**, suggests the Czech koruna is **10.2% undervalued**.

Big Mac Index (Raw) for CZK, 2000-2026

2000-2026

150% log scale



Sources: McDonald's; LSEG Workspace; IMF; Eurostat; LebaneseLira.org; Banque du Liban; *The Economist*

Source: <https://www.economist.com/big-mac-index>

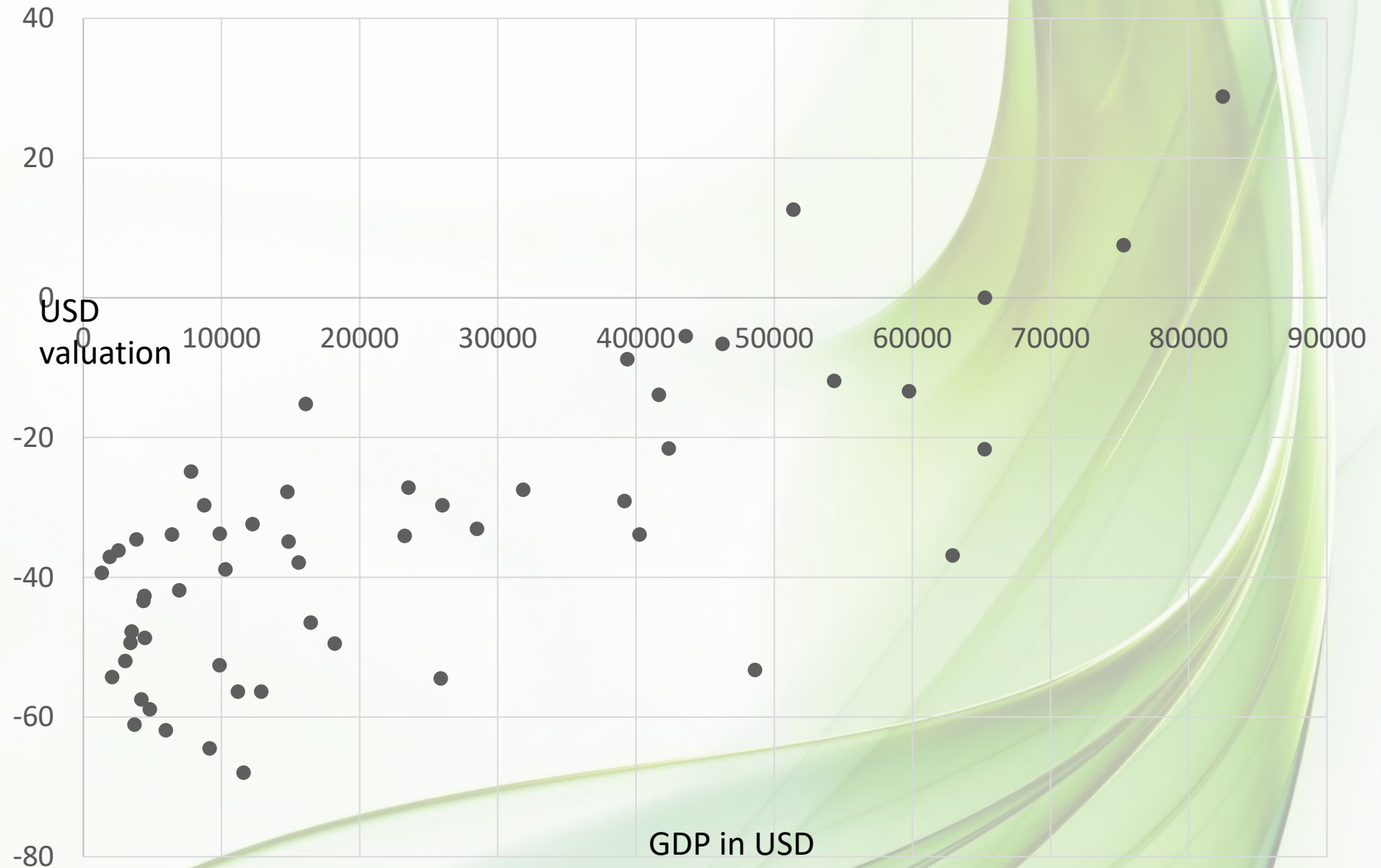
Limitations of the Logic of Purchasing-Power Parity

- Assumptions for arbitrage not met
- Many goods are not easily traded or shipped from one country to another.
 - **Non-tradables**
 - The LOOP obviously does not have to be valid for them, what are the implications for the PPP?
- Tradable goods are **not always perfect substitutes** when they are produced in different countries.
 - What happens if we are always willing to pay 30% higher price for French wine than for German wine?
 - Time and elasticities matter as well!
 - Differences in quality/attractiveness of the goods are not stable over time!
- FX demand/supply reflects also other types of transactions than traditional international trade

Absolute PPP: Empirical Issues

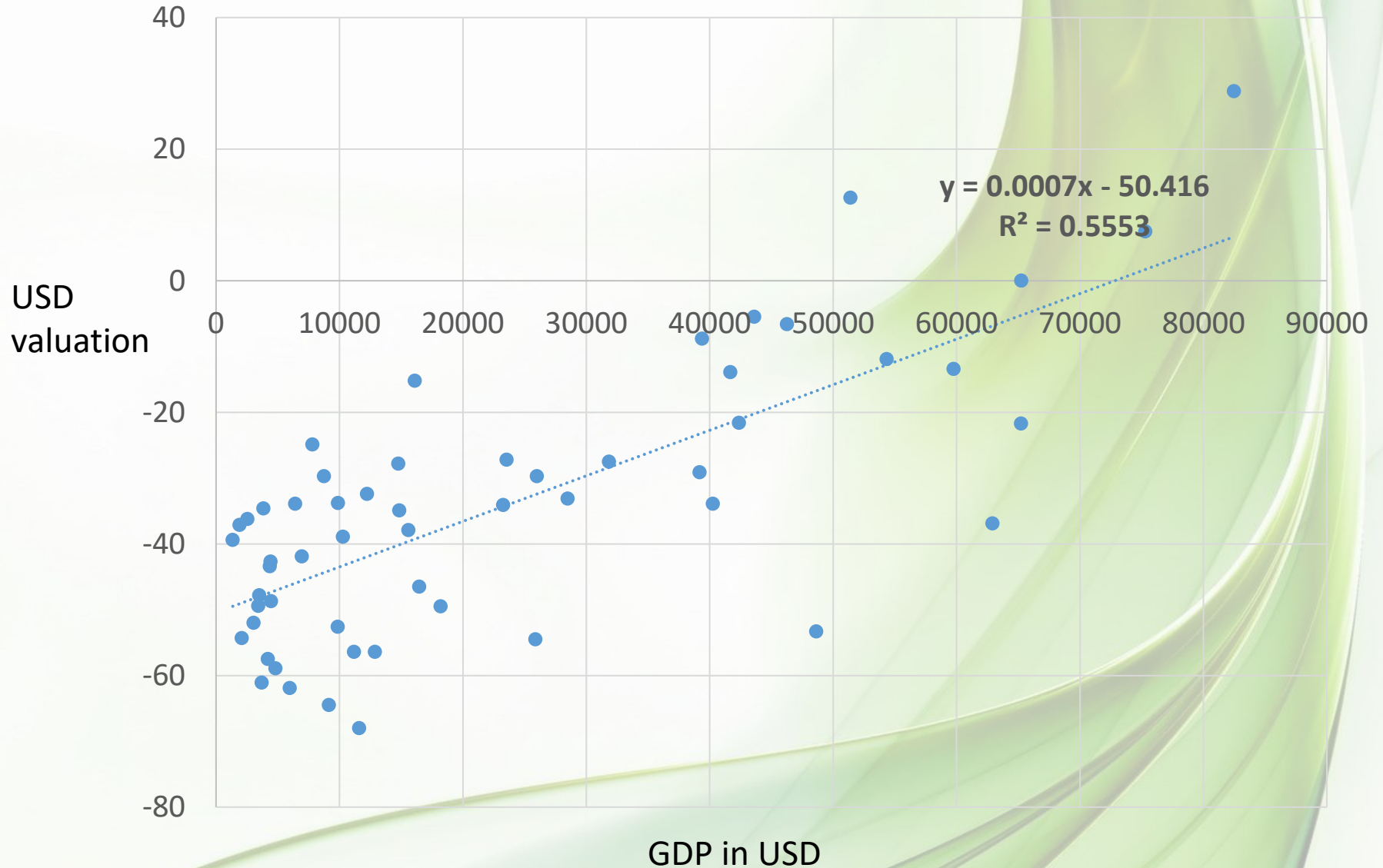
- There not only international but also **intra-country differences in purchasing power....**
 - Especially in prices of so-called non-tradable commodities
 - Examples:
 - Engel & Rogers (1996): price differentials in cities across Canada and the USA
 - And even more importantly: the **discrepancies are systematic** (“puzzle”)

Big Mac Indices and the GDP?



Source of data: Big Mac Index - github

Big Mac Indices and the GDP?



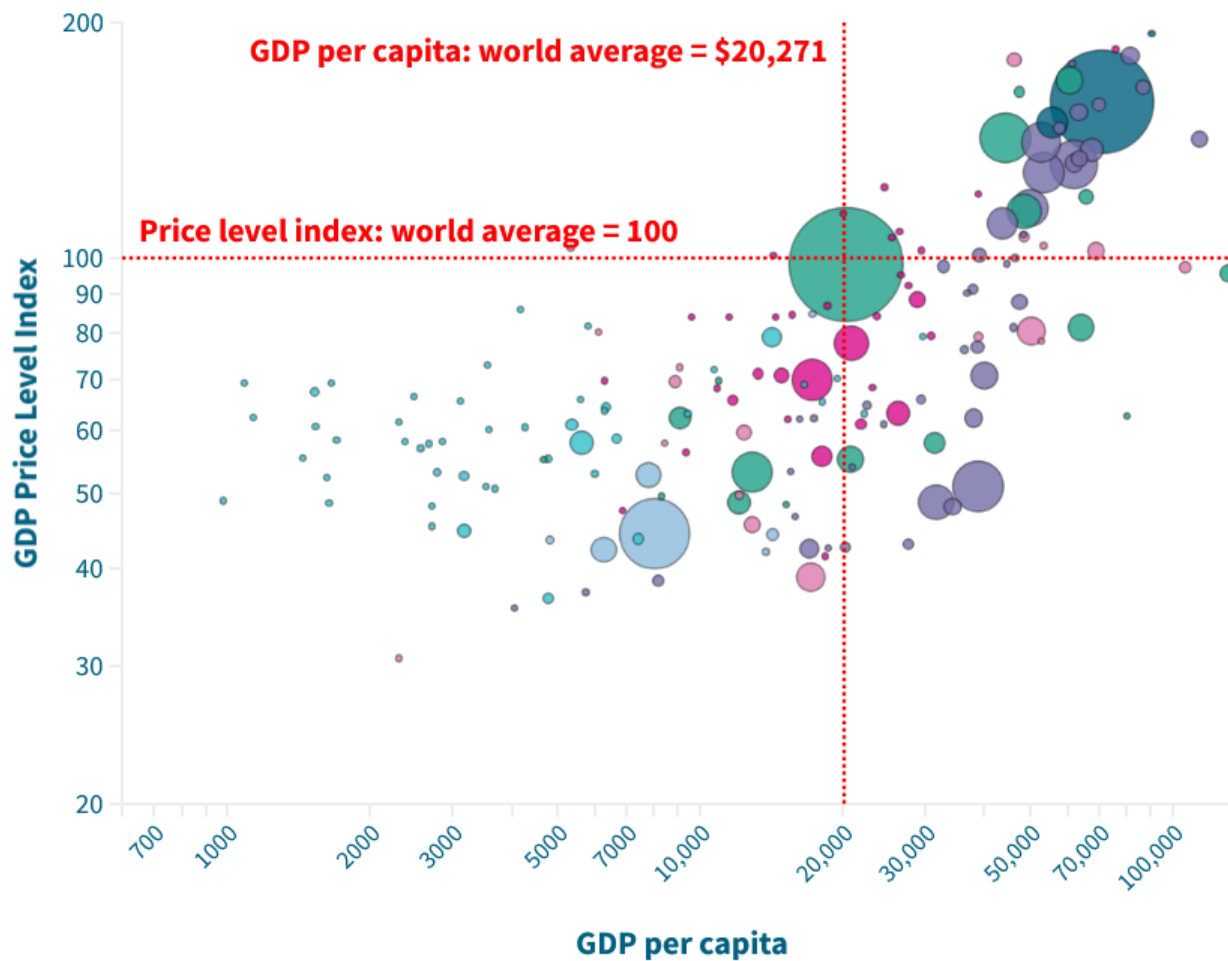
Source of data: Big Mac Index - github

GDP price level index and PPP-based GDP per capita index by economy, 2021

Size of bubble indicates relative size of PPP-based GDP for each economy

Click on chart to view data for each economy

Region ● East Asia & Pacific ● Europe & Central Asia ● Latin America & Caribbean ● Middle East & North Africa
● North America ● South Asia ● Sub-Saharan Africa



Generalized Version of PPP

- What if there are some non-tradables?
- Let's define aggregate price indexes of two countries:

$$P_I = \alpha P_N + (1 - \alpha) P_T$$

$$P_{I*} = \beta P_{N*} + (1 - \beta) P_{T*}$$

- Relative price levels:
$$\frac{P_I}{P_{I*}} = \frac{\alpha P_N + (1 - \alpha) P_T}{\beta P_{N*} + (1 - \beta) P_{T*}}$$

If PPP holds for tradables,
then:

$$S = \frac{P_T}{P_{T*}}$$

$$\frac{P_I}{P_{I*}} = S \cdot \left[\frac{\alpha (P_N / P_T) + (1 - \alpha)}{\beta (P_{N*} / P_{T*}) + (1 - \beta)} \right]$$

- And finally the exchange rate:

$$S = \frac{P_I}{P_{I*}} \cdot \left[\frac{\beta (P_{N*} / P_{T*}) + (1 - \beta)}{\alpha (P_N / P_T) + (1 - \alpha)} \right]$$

Purchasing Power Parity: Relative

- Similar idea as the absolute PPP, but **less restrictive** version
- **We no longer require equal purchasing power at home and abroad**

$$P_{CZ} = x \cdot P_{GER} \cdot S_{CZK/EUR}$$

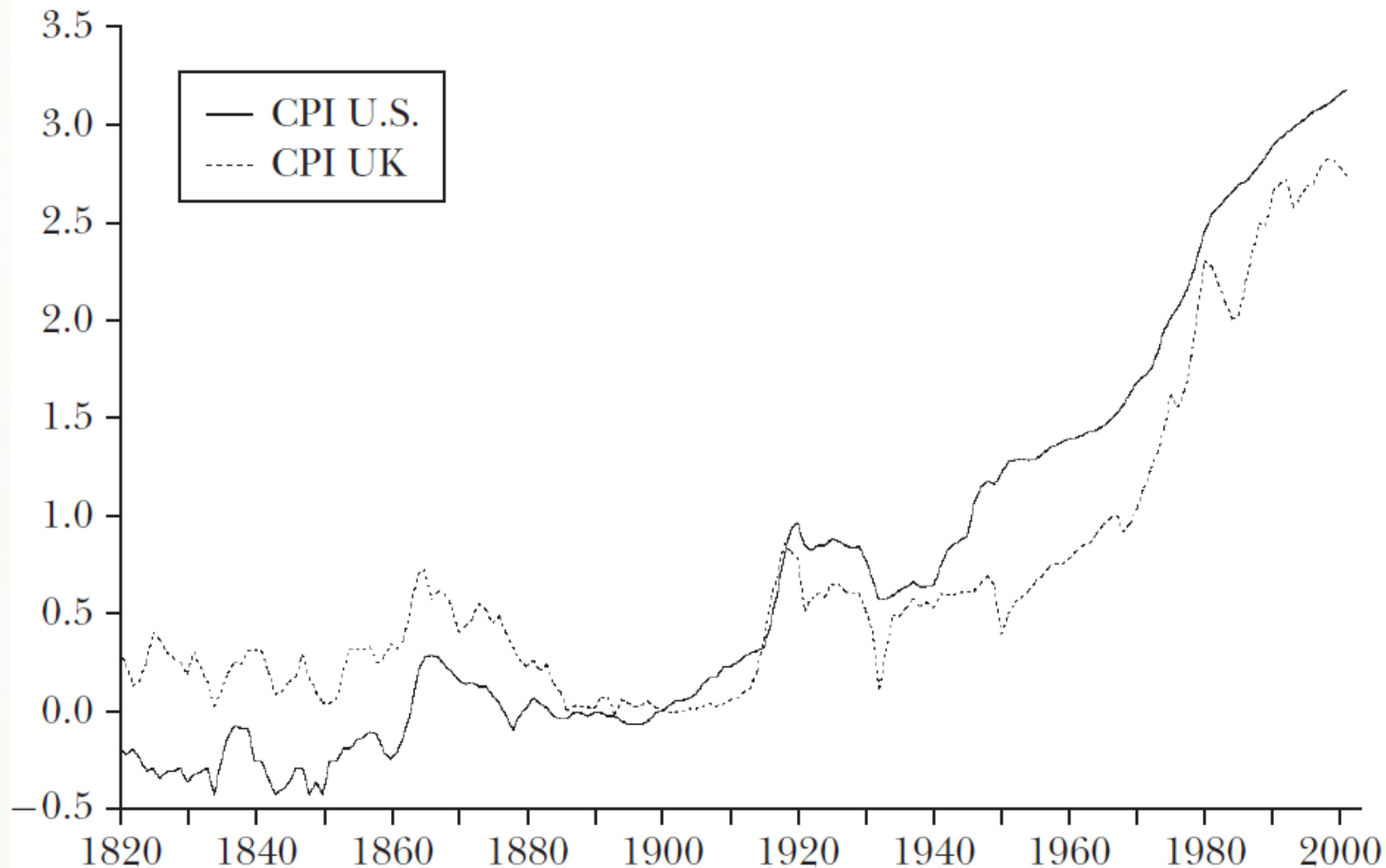
- But we assume that this x is at reasonably stable
- In the long-term the exchange rate should should keep the relative price-competitiveness of the domestic goods stable
- Implications:

$$\Delta S_{CZK/EUR}(\%) = \pi_{CZ} - \pi_{GER}$$

Dollar-Sterling PPP Over Two Centuries

1990 = 0, log scale

(a) U.S. and UK CPIs in dollar terms

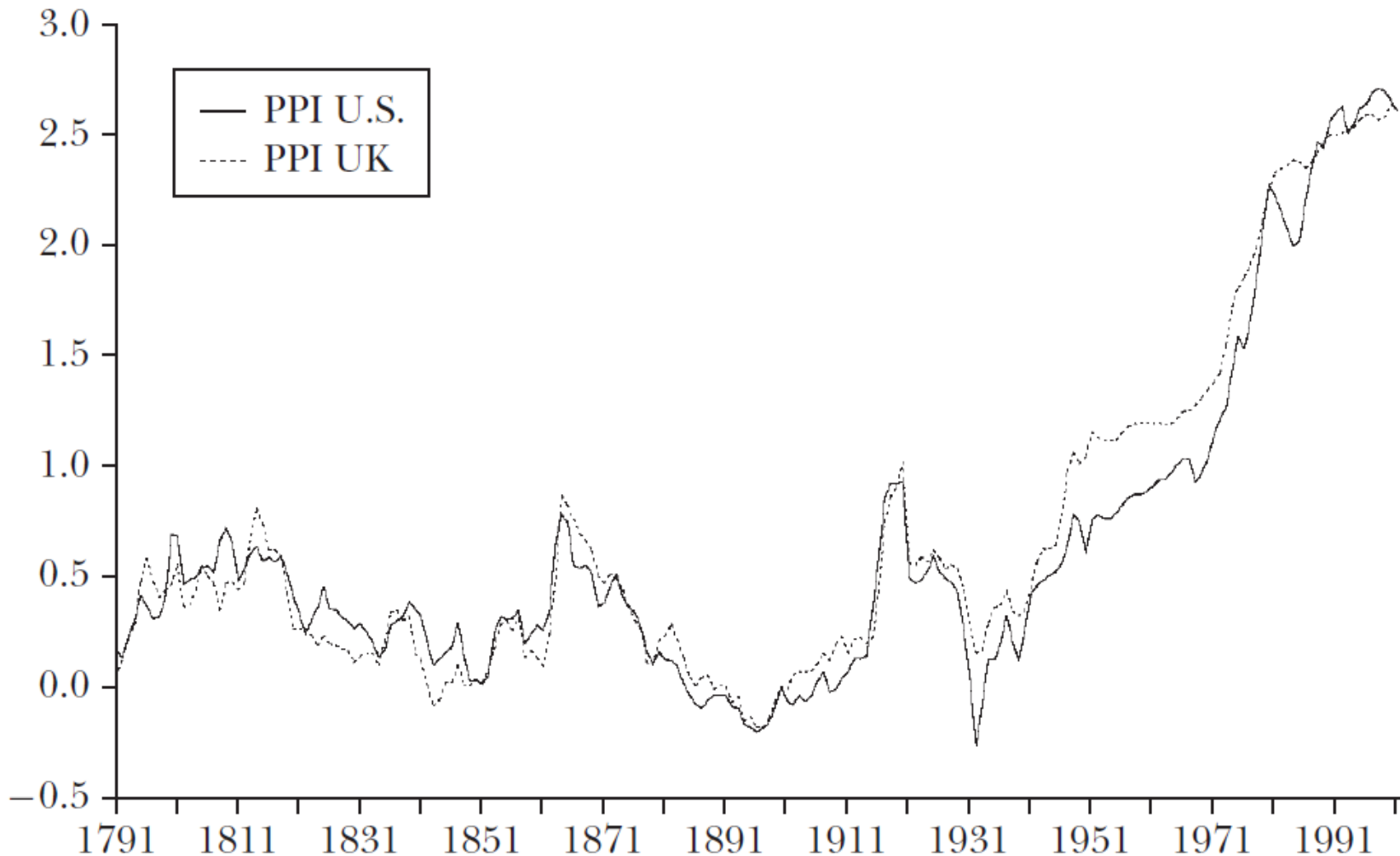


Source: Taylor & Taylor (2004)

Dollar-Sterling PPP Over Two Centuries

1990 = 0, log scale

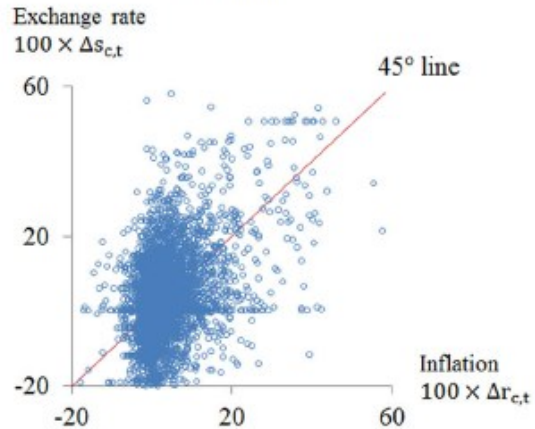
(b) U.S. and UK PPIs in dollar terms



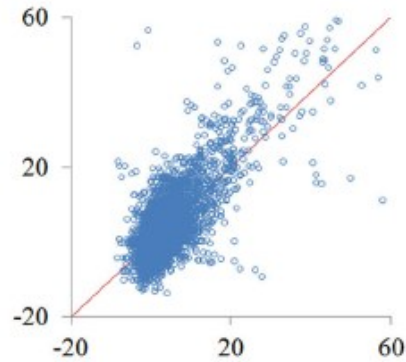
Source: Taylor & Taylor (2004)

PPP at Various Time Horizons

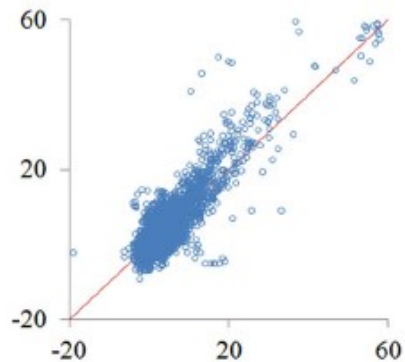
(a) 1-year horizon



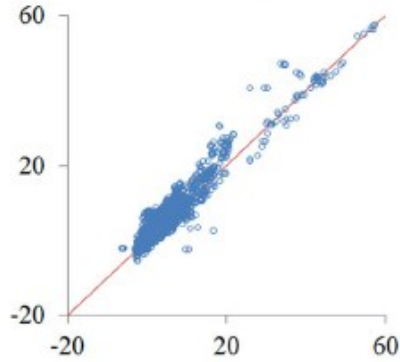
(b) 5-year horizon



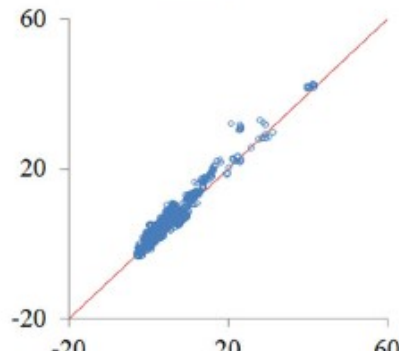
(c) 10-year horizon



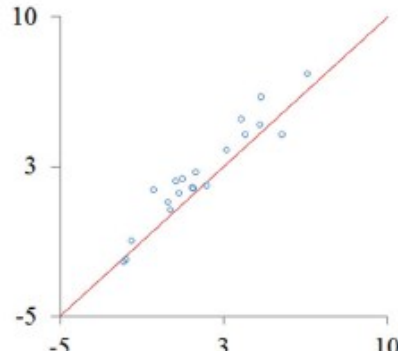
(d) 20-year horizon



(e) 30-year horizon



(f) 44-year horizon



159 countries,
1974-2018

Source: Vo & Vo (2022)

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PPP: Empirical Tests, Indicators & *Additional Issues*

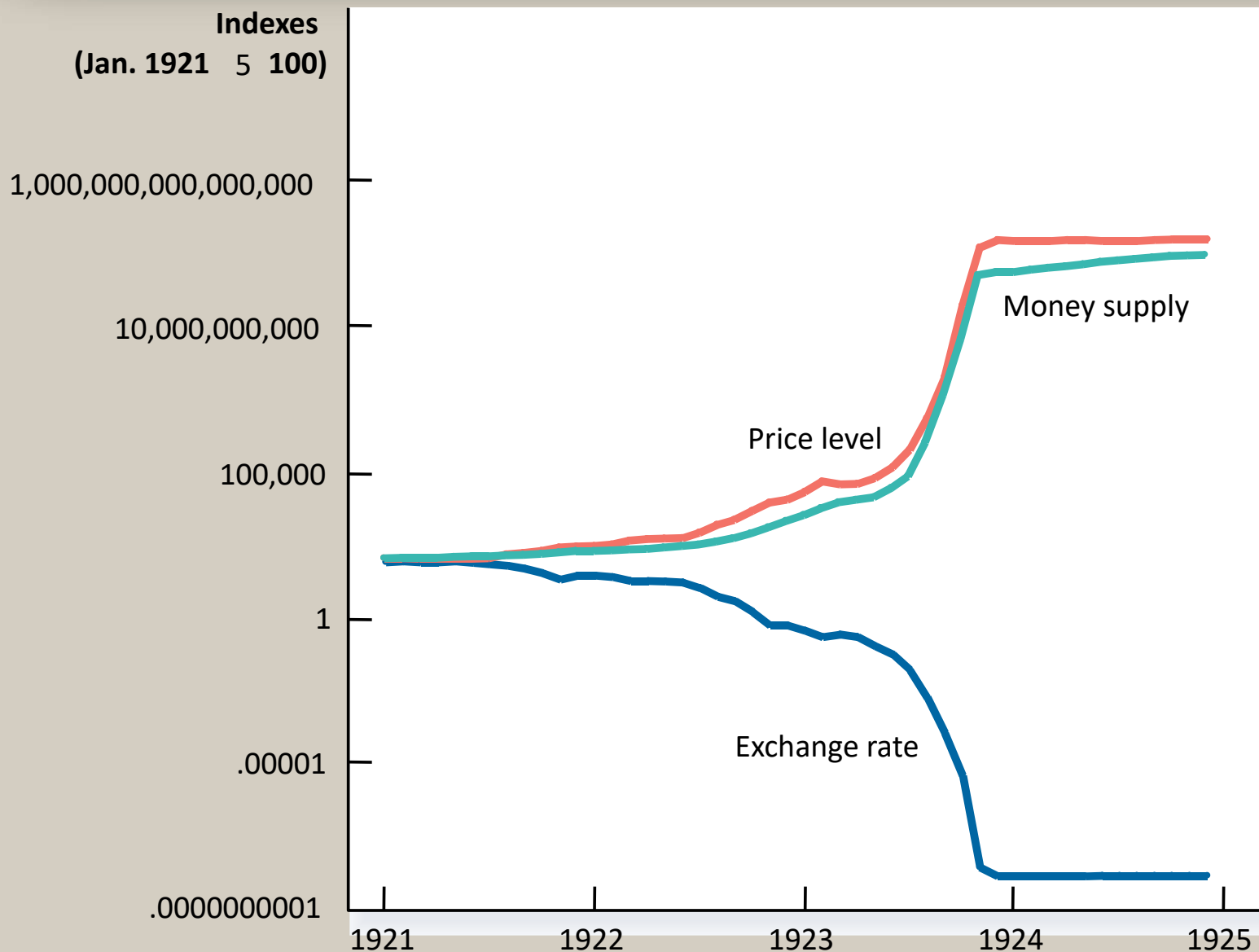
Performance in General

- Systematic issues – at least in short- and medium-run
- Large and volatile SR deviations from the PPP
- LR: real exchange rates might converge to the PPP, but the rate of convergence extremely slow
 - Rogoff: deviations attempt to damp out at 15 % per year

PPP and Hyperinflations

- PPP seems to work relatively well in two cases:
 - Extreme situations (hyperinflations)
 - When the central bank prints large quantities of money, the money loses value both in terms of the goods and services it can buy and in terms of the amount of other currencies it can buy.
 - Long run perspective (but this can be moot)

Money, Prices, and the Nominal Exchange Rate During the German Hyperinflation



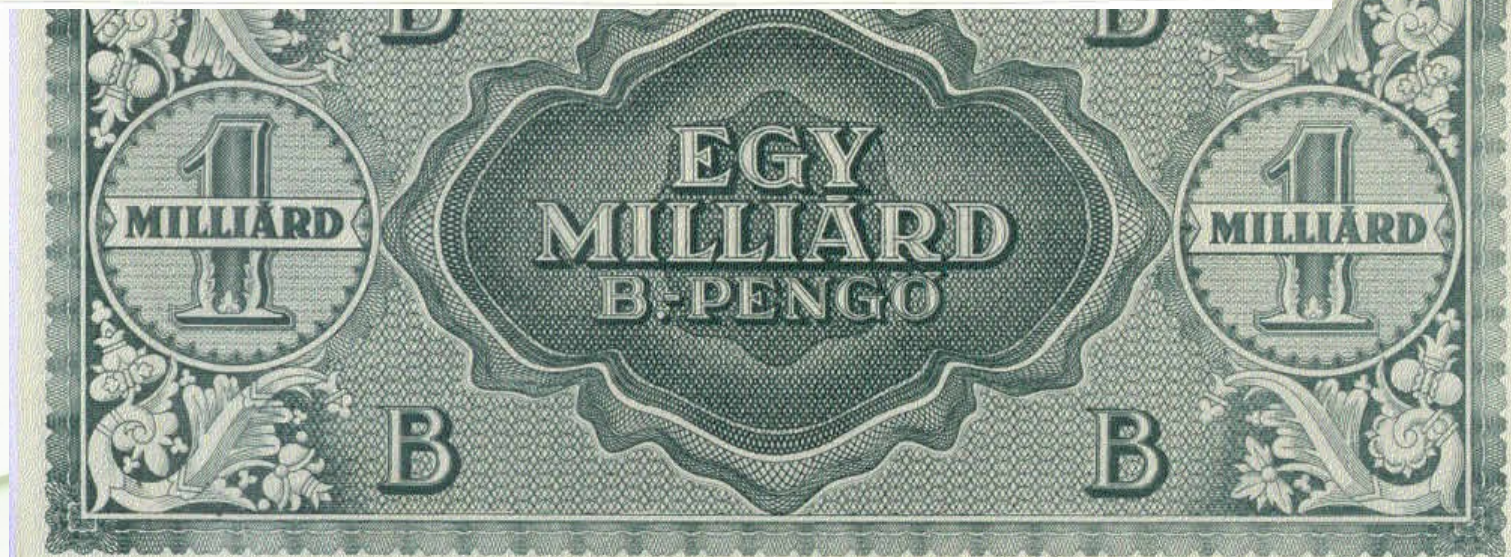
Another Extreme Case: Hungary at the end of WWII

- **Defeated** and occupied by Soviet army
- **Economic collapse**, allegedly partially intentional (liquidation of higher classes by means of printing of banknotes)
- Record growth of prices by **$41.9 \times 10^{16} \%$ a month** (July 1946), i.e. on average prices **doubled every 15 hours**
- Banknotes
 - Highest denomination before 1945: 1 000 Pengö
 - By the end of 1945: 10 000 000 Pengö
 - The highest ever (middle of 1946):
100,000,000,000,000,000,000 Pengö
- You can find additional details in this Business Insider article:
<https://www.businessinsider.com/hungarys-hyperinflation-story-2014-4>

Exchange Rates of Pengö

Date	Pengö per 1 USD
01-01-1927	5.00
31-12-1937	5.40
31-03-1941	5.06
30-06-1944	33.51
31-08-1945	1 320
31-10-1945	8 200
30-11-1945	108 000
31-12-1945	128 000
31-01-1946	795 000
31-03-1946	1 750 000
31-04-1946	59 000 000 000 (5.9×10^{10})
31-05-1946	42 000 000 000 000 000 (4.2×10^{16})
31-07-1946	460 000 000 000 000 000 000 000 000 000 (4.6×10^{29})

Source: wikipedia



Tests of the LOOP & PPP: Strategies

- Cross-country comparison of price data, tests for the presence of systematic biases
 - Problems:
 - Price fluctuations, comparability of individual commodities, identification of sufficiently representative data

Empirical Tests of the LOOP

- One of the most interesting tests:
 - K. Rogoff, K.A. Froot, M. Kim: The Law of One Price Over 700 Years (IMF WP/01/174, 2001)

This paper examines annual commodity price data from England and Holland over a span of seven centuries. Our data incorporates transaction prices on seven commodities: barley, butter, cheese, oats, peas, silver, and wheat, as well as pound/shilling nominal exchange rates going back, in some cases, to 1273. We find that the magnitude, volatility, and persistence of deviations from the law of one price have not declined by as much as one might expect. We find this despite lower transport costs, reduced trade protection, and fewer wars and plagues in the modern era. Our analysis is consistent with growing evidence that goods-market arbitrage remains highly imperfect, even today.

... **goods-market arbitrage remains highly imperfect, even today.**

Tests of the LOOP & PPP: Strategies

- Cross-country comparison of price data, tests for the presence of systematic biases
 - Problems:
 - Price fluctuations, comparability of individual commodities, identification of sufficiently representative data
- Analysis of the properties of **real exchange rates**:
 - Are they **stationary** or not?
 - Gist: get a time series (e.g. real exchange rate CZK/EUR) and use traditional stationarity tests (ADF, Philips-Perron)

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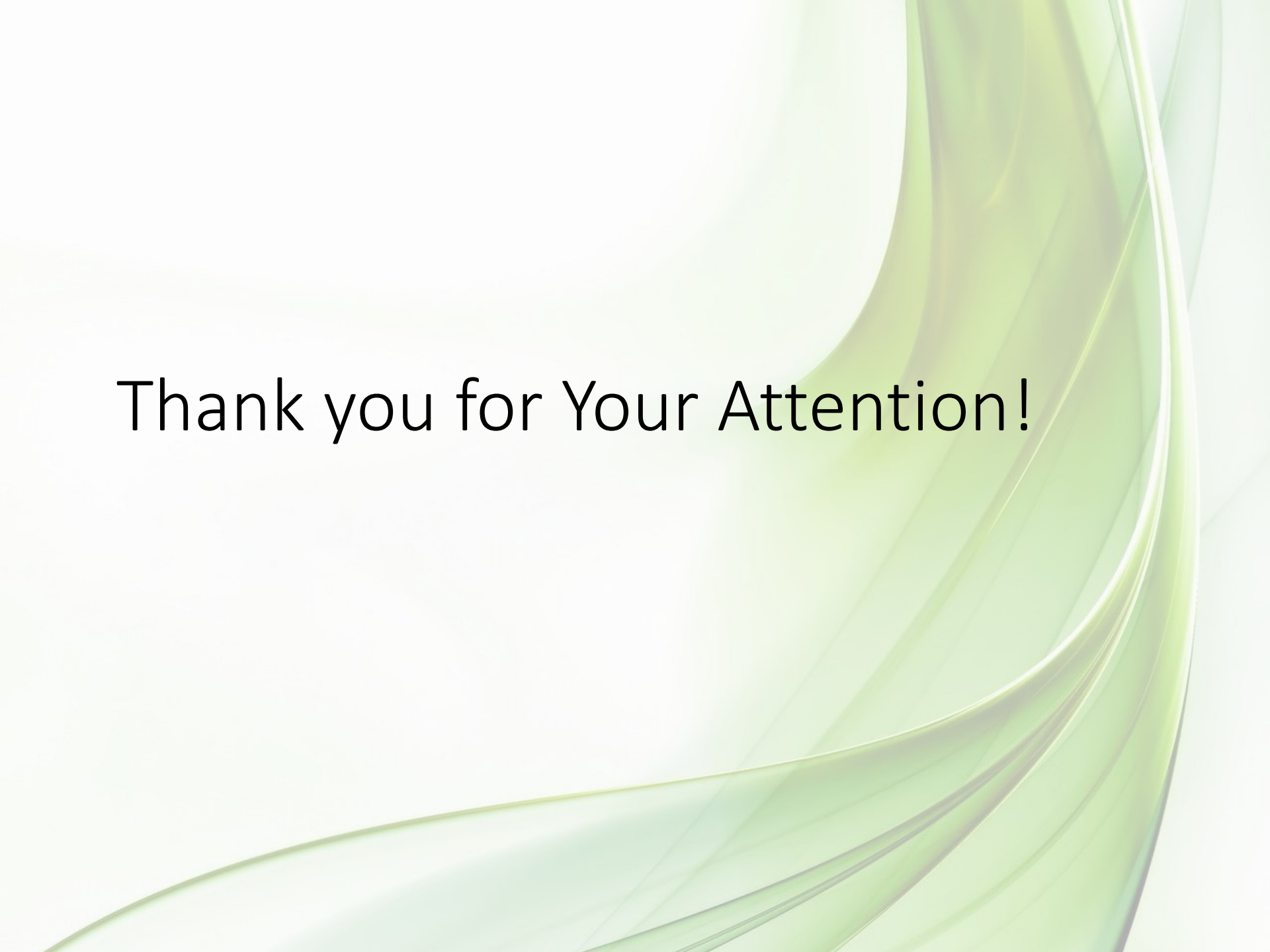
Additional Issues: Models and Forecasting

Exchange Rate Models: Empirical Tests

- Mussa (1979): the spot exchange rate approximately follows **a random-walk process** and **most changes in exchange rates are unexpected**
- **Meese & Rogoff (1983): superiority of the random-walk model in the out-of-sample exchange rate forecasts**
 - None of the structural exchange rate models used in their paper could significantly outperform a simple random-walk model in both short- and medium-terms.
 - Current spot exchange rate as the best prediction?

Exchange Rate Models: Empirical Tests (2)

- But there is some evidence that we can do (sometimes) better than random walk
 - **Lam, Fung, Yu (2008): Comparing Forecast Performance of Exchange Rate Models**
 - Compare PPP, UIP, monetarist sticky price model etc
 - Out-of-sample forecasts
 - Combination of models may lead to better results
 - Depending on the currencies and the forecast horizons, **some of these models outperform common benchmarks** given by the random-walk model and the historical average return.
 - But: No single model consistently stands out as the best exchange-rate forecasting model when assessed by different criteria.
 - **Cheung et al (2019): Exchange rate prediction redux: New models, new data, new currencies**
 - Examine model performance at various forecast horizons (1 quarter, 4 quarters, 20 quarters) using differing metrics (mean squared error, direction of change)
 - The purchasing power parity estimated in the error correction form delivers the best performance at long horizons by a mean squared error measure.
 - Along a direction-of-change dimension, **certain structural models do outperform a random walk with statistical significance.**

The background of the slide features a series of smooth, flowing, translucent green and white lines that curve and swirl across the frame, creating a sense of movement and depth. The lines are most prominent on the right side, where they form a large, sweeping arc that extends towards the bottom left.

Thank you for Your Attention!

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